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GATE ECE · DIGITAL CIRCUITS · CHAPTER-WISE PYQS

Number System and Code Convertions

GATE ECE Digital Circuits · Previous Year Questions with Answer Key

17	15	1987 – 2026	English
QUESTIONS	PAPERS COVERED	YEARS	LANGUAGE

INSIDE THIS PDF

Marks 1	13 questions
Marks 2	4 questions
Answer key	at the end

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Marks 1 13 questions

Q1. GATE ECE 2024

A machine has a 32-bit architecture with 1-word long instructions. It has 24 registers and supports an instruction set of size 40. Each instruction has five distinct fields, namely opcode, two source register identifiers, one destination register identifier, and an immediate value. Assuming that the immediate operand is an unsigned integer, its maximum value is _____.

Open solution

Q2. GATE ECE 2024

In a number system of base r, the equation $x^2 - 12x + 37 = 0$ has $x = 8$ as one of its solutions. The value of r is _____.

Open solution

Q3. GATE ECE 2021

A digital transmission system uses a $(7, 4)$ systematic linear Hamming code for transmitting data over a noisy channel. If three of the message-code word pairs in this code ($m_i; c_i$). Where c_i is the codeword corresponding to the ith message m_i, are known to be $(1100; 0101100)$, $(1110; 0011110)$ and $(0110; 1000110)$, then which of the following is a valid codeword in this code?

(A) 1101001 (B) 1011010 (C) 0110100 (D) 0001011

Open solution

Q4. GATE ECE 2014 Set 2

The number of bytes required to represent the decimal number 1856357 in packed BCD (Binary Coded Decimal) form is _____ .

Open solution

Q5. GATE ECE 2007

X = 01110 and Y = 11001 are two 5-bit binary numbers represented in two's complement format. The sum of X and Y represented in two's complement format using 6 bits is:

(A) 100111 (B) 001000 (C) 000111 (D) 101001

Open solution

Q6. GATE ECE 2005

Decimal 43 in Hexadecimal and BCD number system is respectively

(A) B2, 0100 0011 (B) 2B, 0100 0011 (C) 2B, 0011 0100 (D) B2, 0100 0100

Open solution

Q7. GATE ECE 2004

The range of signed decimal numbers that can be represented by 6-bit 1's complement number is

(A) -31 to +31 (B) -63 to +64 (C) -64 to +63 (D) -32 to +31

Open solution

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Q8. GATE ECE 2002

4-bit 2's complement representation of a decimal number is 1000. The number is

(A) +8 (B) 0 (C) -7 (D) -8

Open solution

Q9. GATE ECE 2001

The 2's complement representation of −17 is

(A) 01110 (B) 101111 (C) 11110 (D) 10001

Open solution

Q10. GATE ECE 1998

An equivalent 2's complement representation of the 2's complement number 1101 is

(A) 110100 (B) 001101 (C) 110111 (D) 111101

Open solution

Q11. GATE ECE 1997

A signed integer has been stored in a byte using the 2's complement format. We wish to store the same integer in a 16-bit word. We should

(A) copy the original byte to the less significant byte of the word and fill the more significant with zeros

(B) copy the original byte to the more significant byte of the word and fill the less significant with zeros

(C) copy the original byte to the less significant byte of the word and make each bit of the more significant byte equal to the most significant bit of the original byte

(D) copy the original byte to the less significant byte as well as the more significant byte of the word

Open solution

Q12. GATE ECE 1993

2's complement representation of a 16-bit number (one sign bit and 15 magnitude bits) is FFFF. Its magnitude in decimal representation is

(A) 0 (B) 1 (C) 32, 767 (D) 65, 535

Open solution

Q13. GATE ECE 1987

The subtraction of a binary number Y from another binary number X, done by adding the 2's complement of Y to X, results in a binary number without overflow. This implies that the result is:

(A) negative and is in normal form

(B) negative and is in 2's complement form

(C) positive and is in normal form

(D) positive and is in 2's complement form

Open solution

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Marks 2 4 questions

Q14. GATE ECE 2026

What is the 10 's complement of $(47)_{10}$?

(A) 52 (B) 53 (C) 54 (D) 55

Open solution

Q15. GATE ECE 2020

P, Q and R are the decimal integers corresponding to the 4-bit binary number 1100 considered in signed magnitude, 1 's complement and 2 's complement representation, respectively. The 6 -bit 2 's complement representation of ($P+Q+R$) is

(A) 111101 (B) 110010 (C) 111001 (D) 110101

Open solution

Q16. GATE ECE 2006

A new Binary Coded Pentary (BCP) number system is proposed in which every digit of a base-5 number is represented by its corresponding 3-bit binary code. For example, the base-5 number 24 will be represented by its BCP code 010100. In this numbering system, the BCP code 100010011001 corresponds to the following number in base-5 system

(A) 423 (B) 1324 (C) 2201 (D) 4231

Open solution

Q17. GATE ECE 2004

11001, 1001 and 111001 correspond to the 2's complement representation of which one of the following sets of number?

(A) 25, 9 and 57 respectively (B) -6, -6 and −6 respectively

(C) -7, -7 and −7 respectively (D) -25, -9 and −57 respectively

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Answer Key

Marks 1

1. **2047**	2. **11**	3. **A**	4. **4**	5. **C**
6. **B**	7. **A**	8. **D**	9. **B**	10. **D**
11. **C**	12. **B**	13. **B**		

Marks 2

14. **B**	15. **D**	16. **D**	17. **C**

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

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